Definite integrals

1 Evaluate the following definite integrals:

a $\int_0^3 (4x+5)\,dx$

b $\int_{-2}^0 (10x+4)\,dx$

c $\int_2^4 (6x+5)\,dx$

d $\int_3^4 (-4x+3)\,dx$

e $\int_{-4}^5 (-8x+3)\,dx$

f $\int_{-10}^{-6} (-5x+8)\,dx$

g $\int_{-2}^0 9x^2\,dx$

h $\int_{-1}^3 9x^2\,dx$

i $\int_{-2}^1 \left(x^2+4\right)dx$

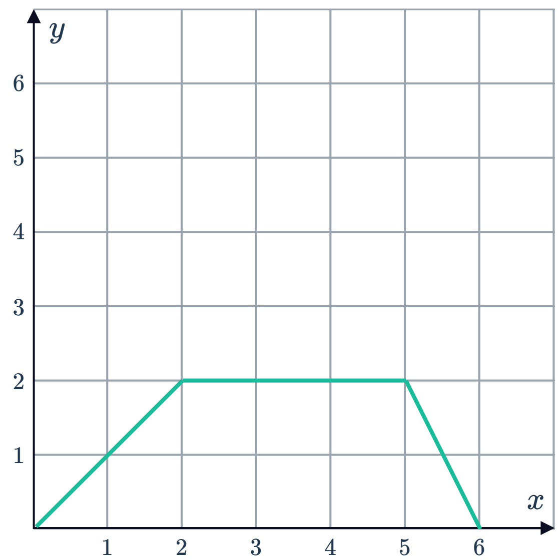
j $\int_{-1}^2 \left(9x^2+1\right)dx$

k $\int_4^6 \left(9x^2+2x+7\right)dx$

l $\int_{-4}^2 x\,(x-4)\,dx$

Area under a curve

2 Consider the graph of $y = f(x)$.



a Find the value of $\int_0^2 f(x)\,dx$.

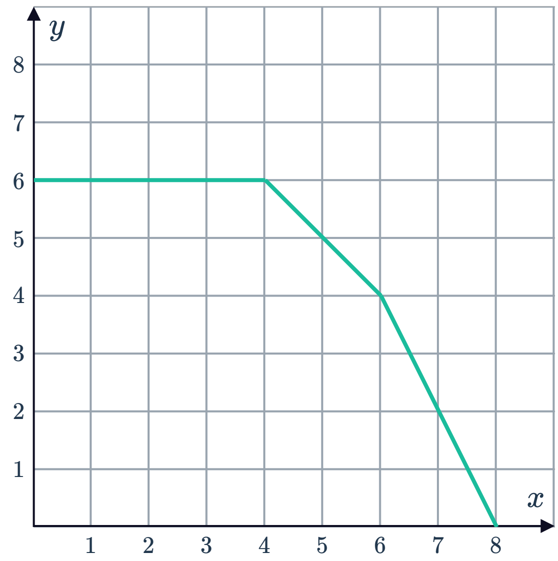
b Find the value of $\int_2^5 f(x)\,dx$.

c Find the value of $\int_5^6 f(x)\,dx$.

d Hence state the area bounded by the function and the x-axis.

e Write a single definite integral to represent the area bounded by the function and the x-axis.

3 Consider the graph of $y = f(x)$.



a Find the value of $\int_0^4 f(x)\,dx$.

b Find the value of $\int_4^6 f(x)\,dx$.

c Find the value of $\int_6^8 f(x)\,dx$.

d Hence state the area bounded by the function and the x-axis.

e Write a single definite integral to represent the area bounded by the function and the x-axis.

4 Consider the function $y = -(x+2)(x+8)$.

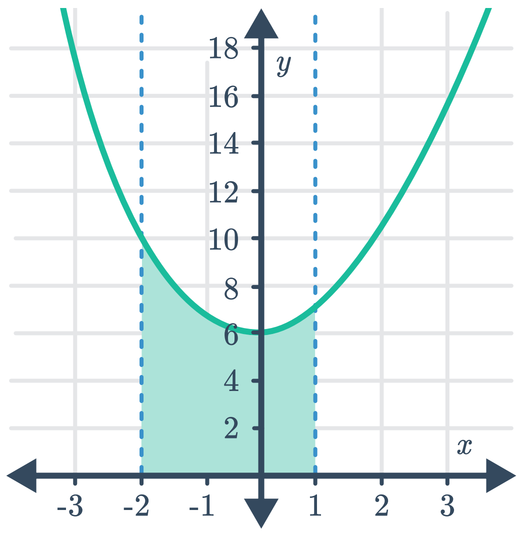
a Find the x-intercepts of the function.

b State the values of x for which the graph is above the x-axis.

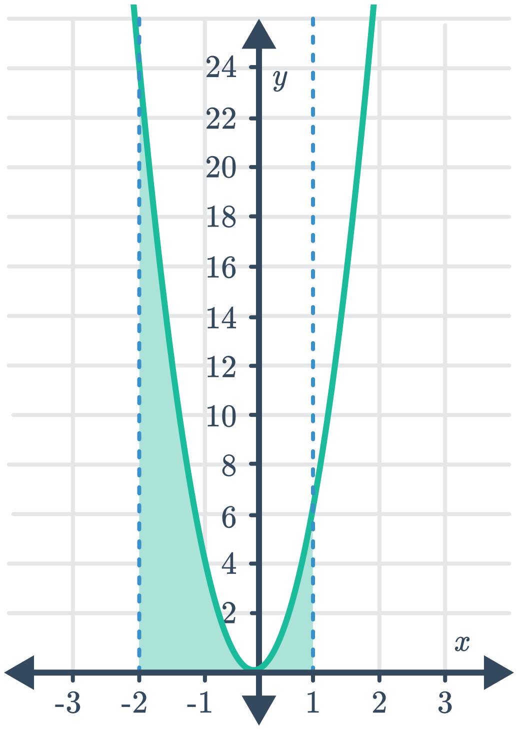
c Calculate $\int_{-8}^{-2} -(x+2)(x+8)\,dx$.

d Hence find the area bounded by the curve, the x-axis and the bounds $x = -8$ and $x = -2$.

5 Consider the graph of the curve $y = x^2 + 6$.
Find the exact area of the shaded region.



6 Consider the graph of the curve $y = 6x^2$.
Find the exact area of the shaded region.



7 Consider the given functions.

i Sketch a graph of the function.

ii Hence calculate the exact area bounded by the curve, $x = 1$, $x = 3$ and the x-axis.

a $y = 2x + 3$ **b** $y = -2x + 8$ **c** $y = 4x - x^2 - 3$